

raft-java System Architecture

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1. Overview

raft-java is a from-scratch Java 17 implementation of the Raft consensus algorithm, following Ongaro & Ousterhout’s “*In Search of an Understandable Consensus Algorithm*” and the PhD dissertation “*Consensus: Bridging Theory and Practice*”.

The project implements:

- **Core Raft** (§5): leader election, log replication, safety rules
- **Leadership transfer** (§3.10): graceful handoff via TimeoutNow RPC
- **Cluster reconfiguration** (§6): one-server-at-a-time membership changes (with §4 errata fix)
- **Log compaction** (§7): independent snapshotting with chunked InstallSnapshot
- **Client interaction** (§8): no-op entry, leader redirection
- **PreVote** (§4.2.3 / §9.6): prevents partitioned servers from disrupting the cluster
- **Performance optimizations** (§10.2): parallel leader disk writes, batching, pipelining
- **Copy-on-write snapshotting** (§5.1): async background compaction
- **Transport abstraction**: pluggable gRPC, Netty TCP, and Hadoop RPC with per-RPC timeouts and TLS/mTLS
- **Storage**: BerkeleyDB JE, append-only WAL, or in-memory (pluggable via `RaftStorage`)
- **Observability**: SLF4J/Logback logging, Micrometer metrics (Prometheus + JMX), health checks, JMX MBean

2. System Architecture

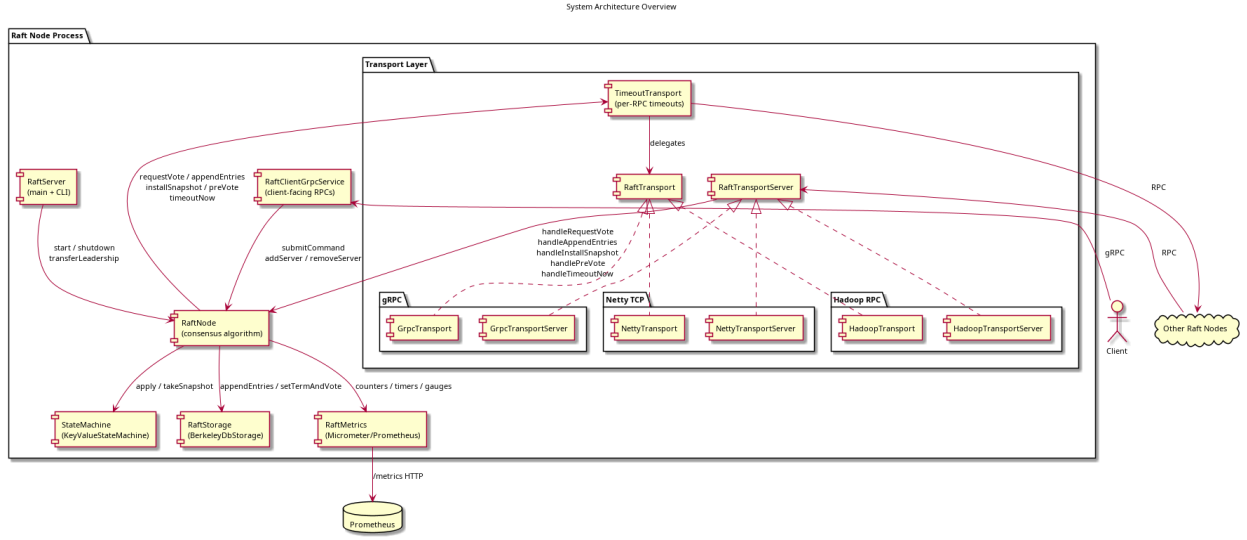


Figure 1: System Architecture Overview

Each Raft node runs as a single JVM process. The **RaftNode** class owns the consensus algorithm. It delegates durable storage to **RaftStorage** (backed by Berkeley DB JE), state machine operations to **StateMachine**, and peer communication to the **Transport** abstraction layer.

The transport layer is pluggable: **gRPC** (default, HTTP/2 + protobuf), **Netty** (raw TCP with custom framing), or **Hadoop RPC** (WritableRpcEngine with BytesWritable-wrapped protobuf). A **TimeoutTransport** decorator applies per-RPC configurable timeouts to any transport implementation.

Logging uses SLF4J with Logback. Prometheus metrics are exposed via Micrometer counters, timers, and gauges on a configurable HTTP endpoint.

3. Class Diagram

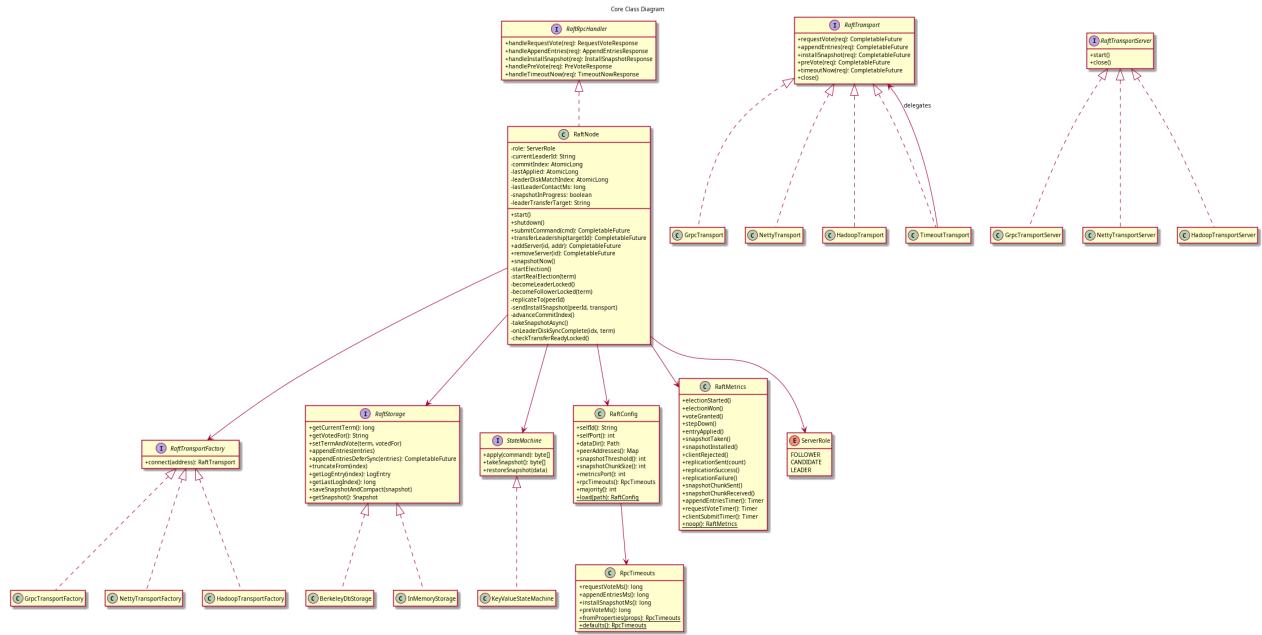


Figure 2: Core Class Relationships

Key design principles:

- **RaftNode** implements **RaftRpcHandler**, making it transport-agnostic
- All dependencies are injected (Spring IoC in production, manual wiring in tests)
- **RaftStorage** separates durable state from the algorithm
- **RaftTransport** / **RaftTransportFactory** abstract the network layer
- **TimeoutTransport** decorates any transport with per-RPC timeouts
- **RaftMetrics** instruments all significant state transitions

4. Leader Election with PreVote

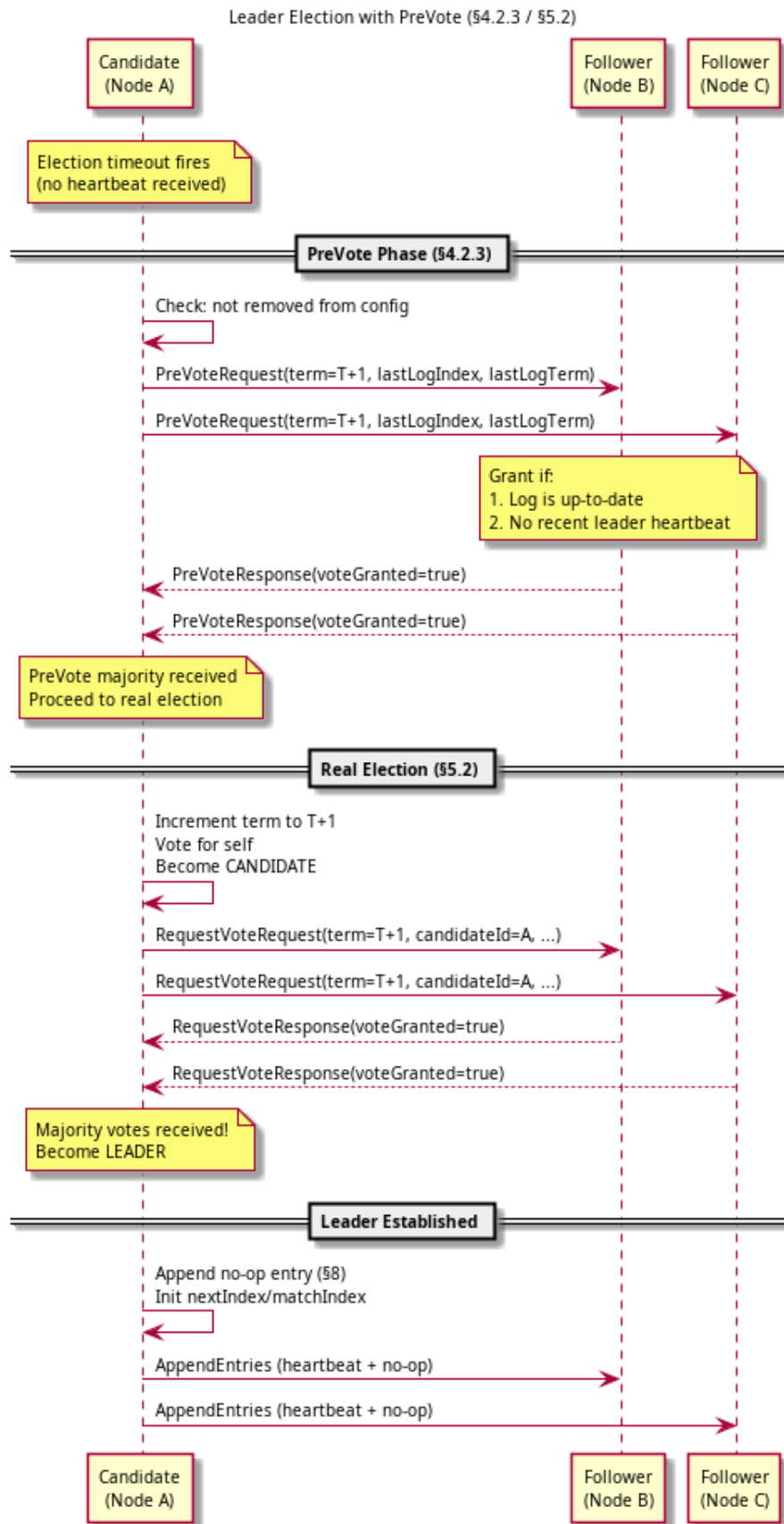


Figure 3: Leader Election Sequence

The election process has two phases:

1. **PreVote (§4.2.3)**: The candidate checks with a majority that they would grant a vote AND haven't heard from a valid leader recently. This prevents a partitioned server from incrementing its term and disrupting the cluster when it rejoins. The term is *not* incremented during PreVote.
2. **Real Election (§5.2)**: Only after PreVote succeeds does the candidate increment its term, vote for itself, and send `RequestVote` RPCs. On receiving a majority, it becomes leader and immediately appends a no-op entry (§8).

Single-node clusters skip PreVote and elect immediately.

5. Leadership Transfer (§3.10)

A leader can gracefully hand off to a specific target server:

1. **Stop accepting client requests** – `submitCommand` returns `NotLeaderException` with the target as leader hint.
2. **Catch up the target** – normal heartbeat replication continues until `matchIndex[target] >= lastLogIndex`.
3. **Send TimeoutNow** – the target immediately starts a real election (skipping PreVote and election timeout), wins, and the prior leader steps down on seeing the higher term.
4. **Abort on timeout** – if transfer doesn't complete within `ELECTION_TIMEOUT_MAX_MS`, the leader aborts and resumes normal operation.

```
node.transferLeadership("n2").get(1, SECONDS);
```

6. Log Replication Pipeline

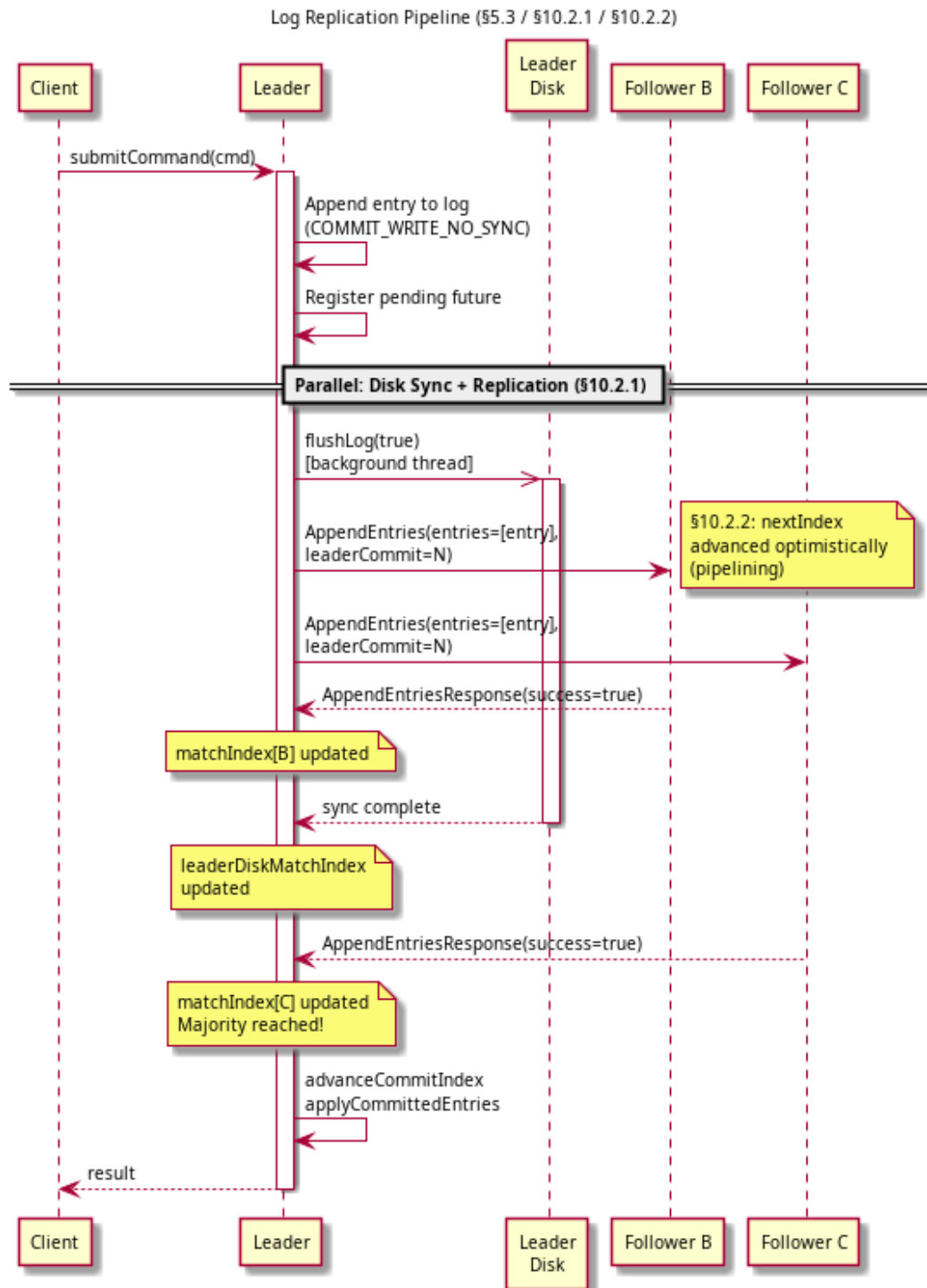


Figure 4: Replication with Parallel Disk Writes and Pipelining

Three optimizations from §10.2 reduce latency:

1. **Parallel leader disk writes (§10.2.1):** The leader writes entries with `COMMIT_WRITE_NO_SYNC` and replicates to followers immediately. A background thread calls `flushLog(true)` to fsync. The leader tracks its own durable progress via `leaderDiskMatchIndex` and only counts itself in the commit majority when the fsync is confirmed.
2. **Batching (§10.2.2):** Each `AppendEntries` RPC carries up to 1 MB of entries, amortizing per-RPC overhead under high load.
3. **Pipelining (§10.2.2):** After sending entries, the leader optimistically advances `nextIndex` so the next heartbeat tick can send subsequent entries without waiting for the previous ACK. A per-peer inflight counter (max 2) prevents unbounded pipelining. On failure, `nextIndex` reverts to the last confirmed `matchIndex + 1`.

7. Snapshot Transfer

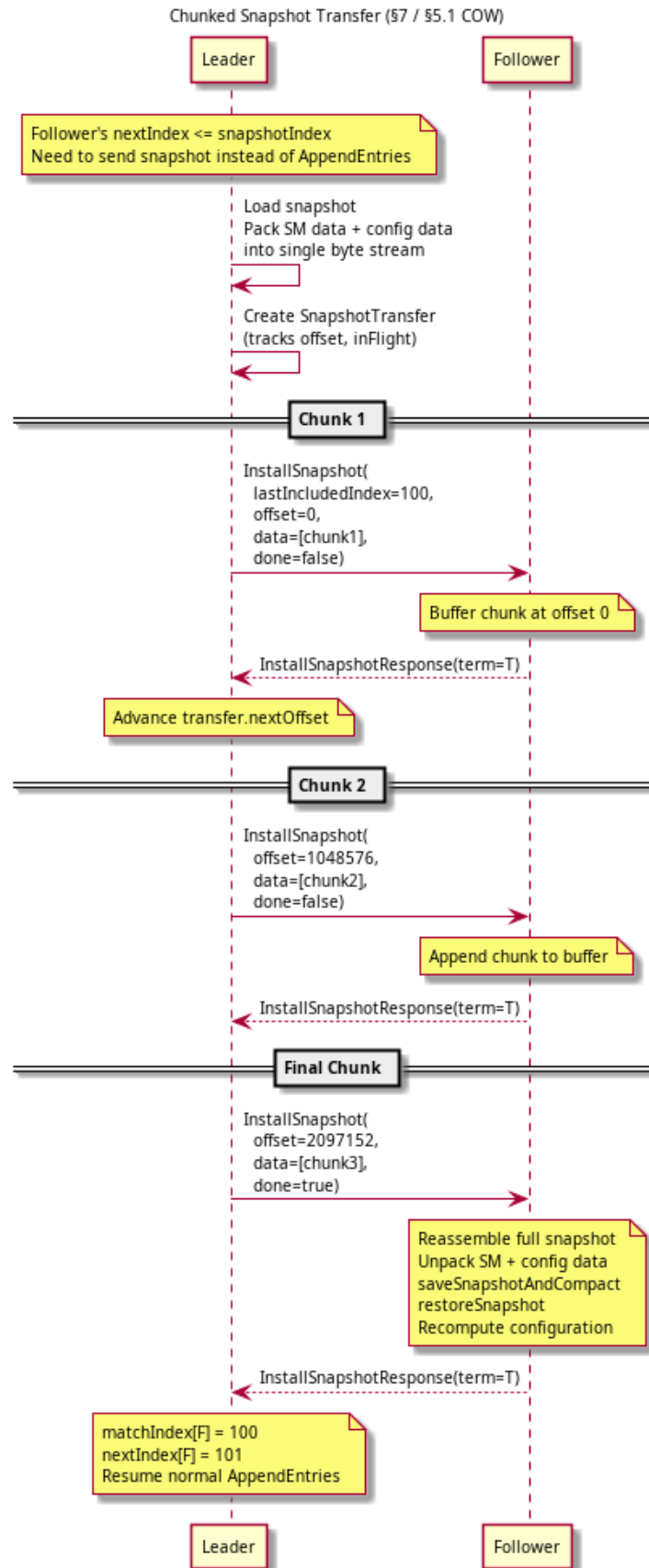


Figure 5: Chunked Snapshot Transfer

Snapshots are handled in two parts:

Taking snapshots (§5.1 COW): Automatic snapshots capture the state machine bytes under the Raft lock (fast, in-memory), then persist and compact the log on a background thread. This avoids blocking replication and elections during disk I/O. A staleness check prevents a background snapshot from overwriting a more recent `InstallSnapshot`.

Transferring snapshots (§7, Figure 13): The leader packs the state machine data and cluster configuration into a single byte stream (`[4B smLen] [smData] [cfgData]`) and sends it in fixed-size chunks (default 1 MB). One chunk is sent per heartbeat cycle. The follower buffers incoming chunks and applies the reassembled snapshot when the final chunk arrives (`done=true`).

8. Transport Layer

The transport abstraction decouples `RaftNode` from any specific network library:

Interface	Purpose
<code>RaftTransport</code>	Outbound peer connection (5 RPCs)
<code>RaftTransportFactory</code>	Creates outbound connections from an address string
<code>RaftTransportServer</code>	Inbound server that routes RPCs to <code>RaftRpcHandler</code>
<code>RaftRpcHandler</code>	Handler interface (implemented by <code>RaftNode</code>)
<code>RpcTimeouts</code>	Per-RPC timeout configuration from <code>.properties</code>
<code>TimeoutTransport</code>	Decorator applying <code>orTimeout()</code> to any transport
<code>TlsConfig</code>	TLS/mTLS certificate paths and settings

RPCs: `RequestVote`, `AppendEntries`, `InstallSnapshot`, `PreVote`, `TimeoutNow`.

Implementations:

Transport	Protocol	Use case
<code>gRPC</code> (default)	HTTP/2 + protobuf	Standard deployments
Netty TCP	Length-delimited + protobuf	Lightweight, no HTTP overhead
Hadoop RPC	<code>WritableRpcEngine</code> + <code>BytesWritable</code>	Hadoop ecosystem integration

Per-RPC timeouts are configurable via `.properties`:

```
rpc.timeout.request.vote.ms=1000
rpc.timeout.append.entries.ms=2000
rpc.timeout.install.snapshot.ms=30000
rpc.timeout.pre.vote.ms=1000
```

TLS/mTLS: `gRPC` and Netty transports support TLS encryption and mutual authentication via `TlsConfig` (parsed from `.properties`). `gRPC` uses `GrpcSslContexts` with shaded Netty; raw Netty uses `SslContextBuilder`.

9. Storage Layer

Three `RaftStorage` implementations are available:

Implementation	Durability	Use case
<code>BerkeleyDbStorage</code>	<code>fsync</code> (<code>COMMIT_SYNC</code>)	Production default

Implementation	Durability	Use case
<code>WalStorage</code>	Append-only WAL + fsync	Lightweight alternative
<code>InMemoryStorage</code>	None	Tests and demos

WalStorage uses an append-only file (`wal.log`) with length-prefixed protobuf entries and an in-memory index (log index to file offset) rebuilt on recovery. Metadata (term/vote) and snapshots are stored in separate files, atomically replaced via rename. Supports deferred fsync (§10.2.1) via a background thread. Compaction rewrites the WAL after a snapshot discards the prefix.

10. Errata Fix (§4)

The dissertation’s single-server membership change algorithm has a known safety bug: competing configuration changes across term boundaries in even-sized clusters can use non-overlapping majorities. The fix (from the raft-dev mailing list): a leader must commit an entry from its current term before accepting configuration changes. The no-op entry appended at the start of every term (§8) serves this purpose – `addServer/removeServer` are rejected until `commitIndex >= leaderNoOpIndex`.

11. Observability

Metrics: All significant Raft events are instrumented via Micrometer, published to both **Prometheus** (`/metrics` HTTP endpoint) and **JMX** (via `JmxMeterRegistry` and `CompositeMeterRegistry`):

JMX MBean: `RaftNodeMXBean` (`com.fbtberger.raft:type=RaftNode`) exposes role, term, `commitIndex`, cluster members, and operations (`triggerSnapshot`, `transferLeadership`) for JConsole/VisualVM.

Health checks on the metrics port:

- GET `/health` — liveness, always 200
- GET `/ready` — readiness, 200 when leader is known, 503 during elections

Metrics table:

Metric	Type	Description
<code>raft.election.started</code>	Counter	Elections initiated
<code>raft.election.won</code>	Counter	Elections won
<code>raft.vote.granted</code>	Counter	Votes granted to candidates
<code>raft.stepdown</code>	Counter	Step-downs to follower
<code>raft.entry.applied</code>	Counter	Entries applied to state machine
<code>raft.replication.sent</code>	Counter	Entries sent to followers
<code>raft.replication.success</code>	Counter	Successful <code>AppendEntries</code> responses
<code>raft.replication.failure</code>	Counter	Failed <code>AppendEntries</code> (log mismatch)
<code>raft.snapshot.taken</code>	Counter	Snapshots taken
<code>raft.snapshot.installed</code>	Counter	Snapshots installed from leader
<code>raft.snapshot.chunk.sent</code>	Counter	Snapshot chunks sent
<code>raft.snapshot.chunk.received</code>	Counter	Snapshot chunks received

Metric	Type	Description
<code>raft.client.rejected</code>	Counter	Client submissions rejected (not leader)
<code>raft.rpc.append.entries</code>	Timer	AppendEntries RPC handling time
<code>raft.rpc.request.vote</code>	Timer	RequestVote RPC handling time
<code>raft.rpc.install.snapshot</code>	Timer	InstallSnapshot RPC handling time
<code>raft.client.submit</code>	Timer	Client submit-to-commit time
<code>raft.node.term</code>	Gauge	Current Raft term
<code>raft.node.commit.index</code>	Gauge	Highest committed log index
<code>raft.node.last.applied</code>	Gauge	Last applied log index
<code>raft.node.role</code>	Gauge	Current role (0=follower, 1=candidate, 2=leader)
<code>raft.cluster.size</code>	Gauge	Number of nodes in configuration